3. Implement a preprocessing pipeline using Pipeline and ColumnTransformer

```
import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

data = pd.DataFrame({
    "age": [25, 30, 45],
    "salary": [40000, 50000, 60000],
    "city": ["Mumbai", "Pune", "Delhi"]
})

num_cols = ["age", "salary"]
cat_cols = ["city"]

preprocess = ColumnTransformer([
    ("num", StandardScaler(), num_cols),
    ("cat", OneHotEncoder(), cat_cols)
])

pipeline = Pipeline([
    ("preprocessing", preprocess)
])
output = pipeline.fit_transform(data)
print(output)
```

4. Implement data preprocessing using ColumnTransformer for mixed data types.

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
import pandas as pd

df = pd.DataFrame({
    "height": [150, 160, 170],
    "weight": [50, 60, 70],
    "gender": ["M", "F", "M"]
})

ct = ColumnTransformer([
    ("scale", StandardScaler(), ["height", "weight"]),
    ("encode", OneHotEncoder(), ["gender"])
])

result = ct.fit_transform(df)
print(result)
```

5. Perform complete preprocessing and show step-by-step transformation

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, OneHotEncoder

df = pd.DataFrame({
    "age": [20, 30, 40],
    "city": ["Pune", "Mumbai", "Delhi"]
})

# Step 1: Encode categorical data
encoder = OneHotEncoder()
city_encoded = encoder.fit_transform(df[["city"]]).toarray()

# Step 2: Scale numerical data
scaler = StandardScaler()
age_scaled = scaler.fit_transform(df[["age"]])

print("Encoded City:\n", city_encoded)
print("Scaled Age:\n", age_scaled)
```

6. Program to remove outliers using IQR method

```
import pandas as pd

data = pd.DataFrame({
    "marks": [35, 40, 42, 45, 90, 100]
})

Q1 = data["marks"].quantile(0.25)
Q3 = data["marks"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

clean_data = data[(data["marks"] >= lower) & (data["marks"] <= upper)]

print(clean_data)
```